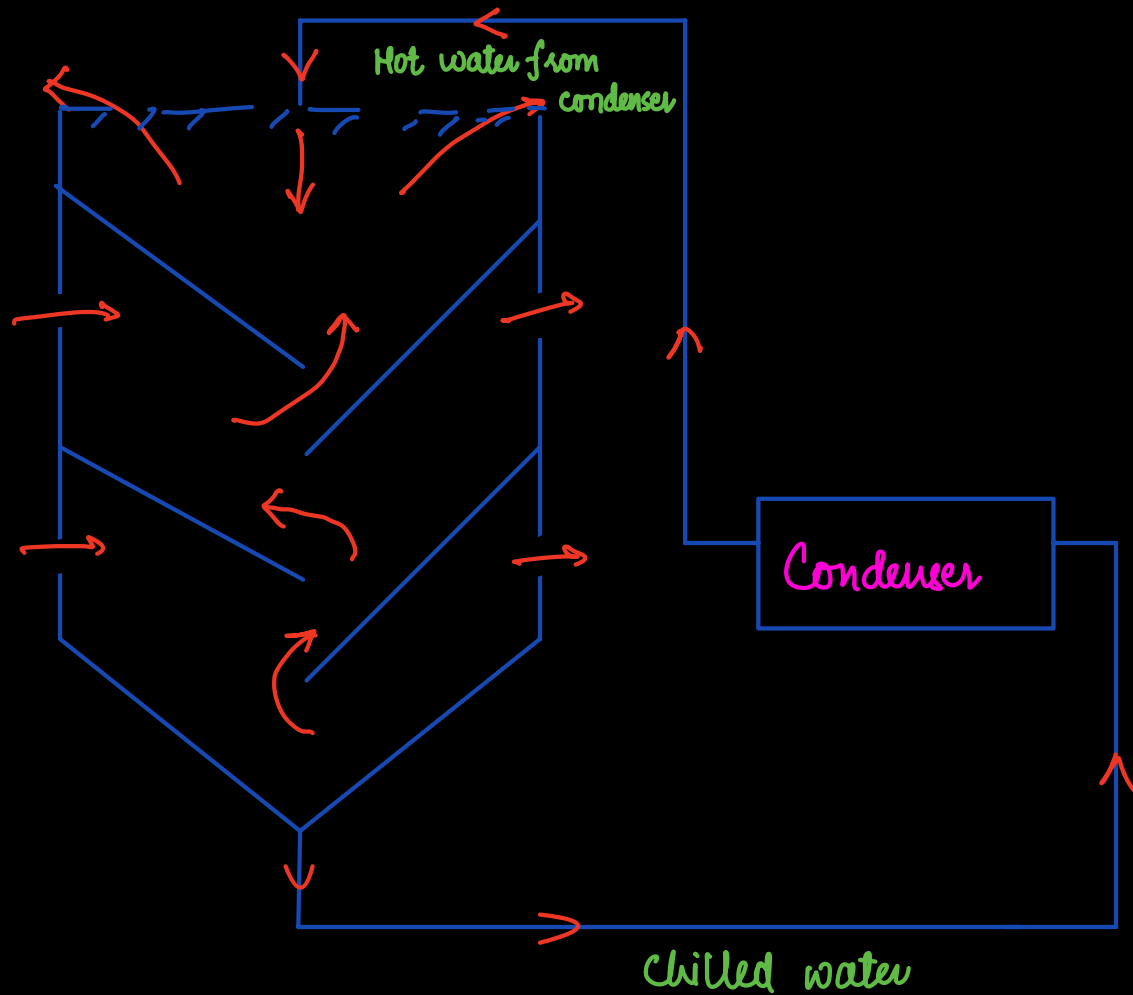


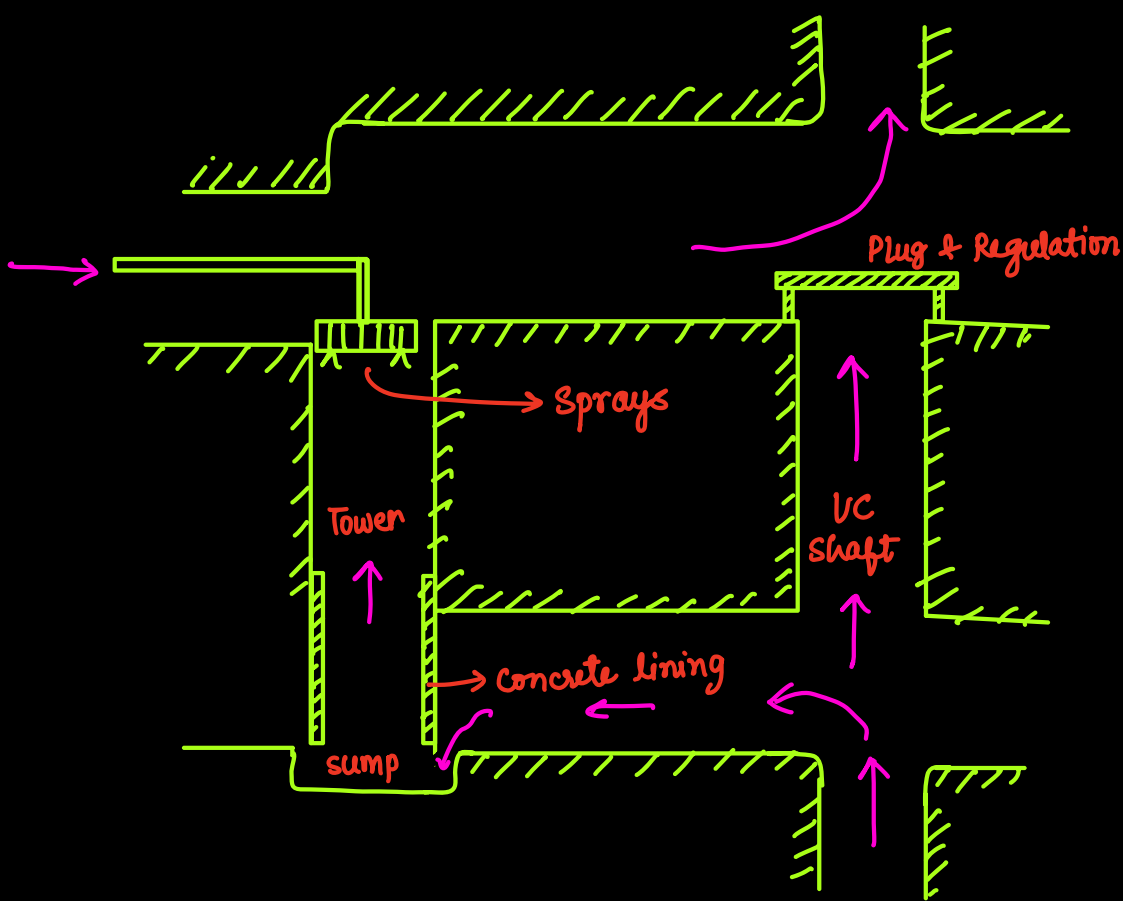
Towers/Spraying Chambers —

Types of spray towers :—

① Natural draught towers

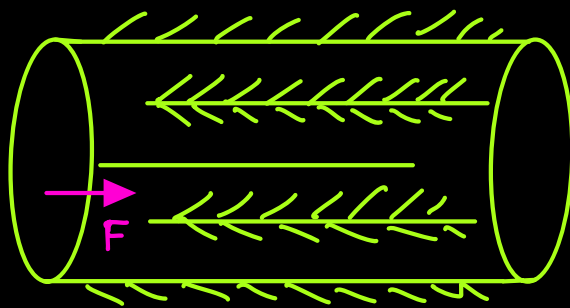
② Mechanical draught towers / counter flow towers





Layout of a typical V/G cooling tower

★ Airflow through mine openings ———



$$\text{Shear stress} = \frac{F}{A}$$

Types of Fluids :-

- (1) Ideal or non-viscous fluid
- (2) Real or viscous fluid

molecular cohesion

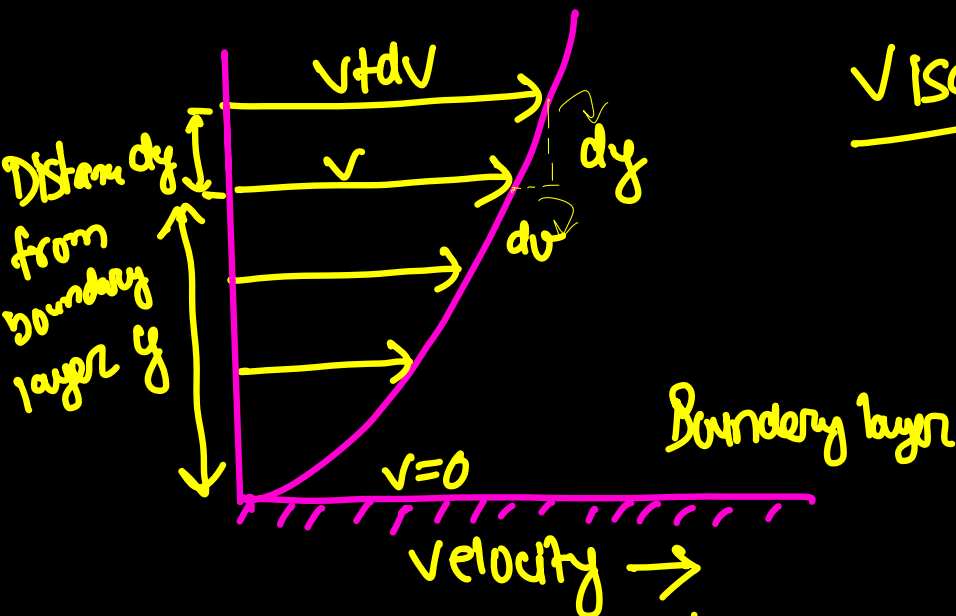
Based on compressibility —

- ① Compressible fluid
- ② Incompressible fluid

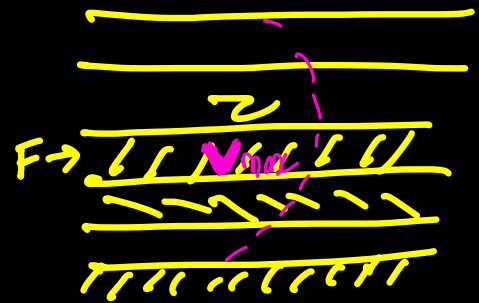
6/3/24

Air = incompressible fluid in MV to use fluid dynamics

Flow of Viscous Fluids: (Real)



Viscous Fluid



$$Q = AV$$

Flow of fluid in an open channel

Tangential shear strain $\propto$ shear strain

$$\tau \propto \frac{dv}{dy}$$

$$\tau = \mu \cdot \frac{dv}{dy}$$

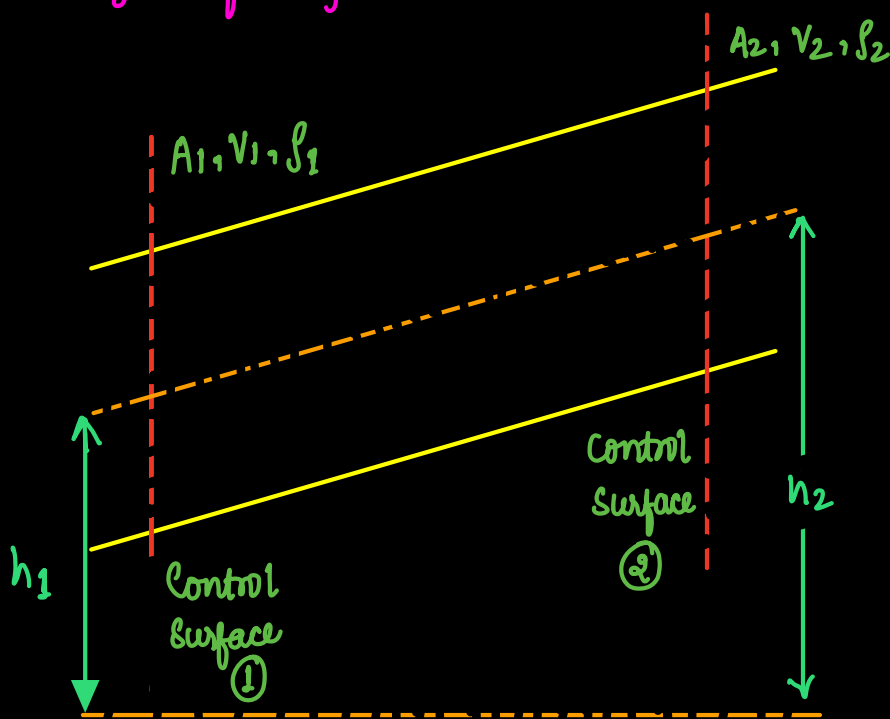
$$\frac{dv}{dy} = \text{velocity gradient (s}^{-1}\text{)}$$

μ = Absolute / Dynamic viscosity, $\frac{N \cdot s}{m^2} = \frac{P \cdot s}{kg}$

$$\nu = \text{kinetic viscosity} \\ = \frac{\mu}{\rho}, \text{m}^2/\text{s}$$

$$\text{In CGS,} \\ \text{Poise (g/cm.s)} \\ = 0.1 \text{ Pa.s}$$

* Analysis of airflow ———



Bernoulli's Equation

- 1.) Continuity equation / conservation of mass
- 2.) Energy equation
- 3.) Momentum equation

* Mass flow rate of air entering the control surface (1) = Mass flow rate leaving the control surface (2)

$$M_1 = M_2, \text{ (kg/s)}$$

$$\Rightarrow P_1 Q_1 = P_2 Q_2$$

$$\text{If } \rho_1 = \rho_2 ;$$

$$\Rightarrow Q_1 = Q_2$$

Bernoulli's Equation —

Considering a unit of mass of fluid flowing in a duct.

$$\text{Total head (m)} = \frac{P}{\rho g} + \frac{v^2}{2g} + h$$

In terms of pressure (energy per unit volume) : —

$$\text{Total pressure (Pa)} = P + \frac{\rho v^2}{2} + \rho g h$$

In case of a real (viscous) fluid : —

$$P_1 + \frac{\rho_1 v_1^2}{2} + \rho_1 \cdot g \cdot h_1 = P_2 + \frac{\rho_2 v_2^2}{2} + \rho_2 \cdot g \cdot h_2$$

For a horizontal duct or opening of uniform cross-section

$$h_1 = h_2$$

$$Q_1 = Q_2$$

$$v_1 = v_2$$

$$\rho_1 = \rho_2$$

$$P_1 = P_2$$

In case of a real (viscous) fluid : —

$$P_1 + \frac{\rho_1 v_1^2}{2} + \rho_1 \cdot g \cdot h_1 = P_2 + \frac{\rho_2 v_2^2}{2} + \rho_2 \cdot g \cdot h_2 + F_{1-2}$$

$F_{1-2} \Rightarrow$ Frictional loss P

For a horizontal duct or opening of uniform cross-section

$$h_1 \neq h_2$$

$$Q_1 \neq Q_2$$

$$v_1 \neq v_2$$

$$\rho_1 \neq \rho_2$$

$$P_1 \neq P_2$$

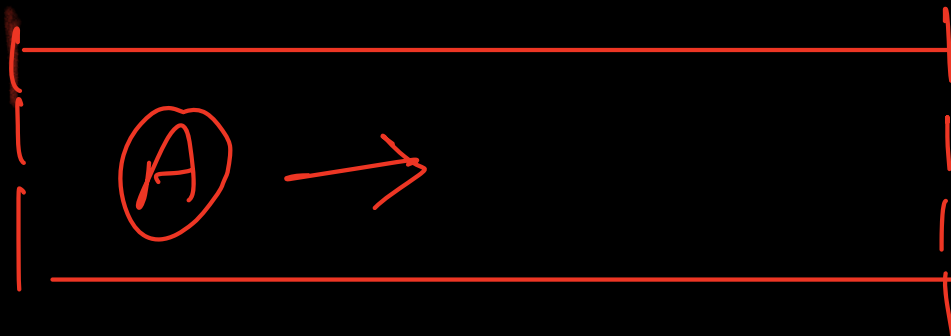
$$P_1 - P_2 = F_{1-2}$$

Depending upon the Momentum equation (conservation of momentum) the sum of external forces acting on the control surface in a given direction is equal to the rate of change of air momentum in that direction.

Airflow in Airways

Inlet

Outlet



$$P_1 = P_2 + F_{1-2}$$

or

$$P_1 - P_2 = F_{1-2}$$

Pressure loss in Airways :

> In case of laminar flow of fluid (viscous) fluid
(Hagen-Poiseuille's Equation)

$$\Delta P = \frac{32 \mu \cdot L \cdot v}{D^2}$$

μ = Dynamic Vis. (Pa.s)

L = length of duct

v = average velocity

D = Diameter of Duct

$$D_e = \frac{4A}{P}$$

> In Turbulent flow:

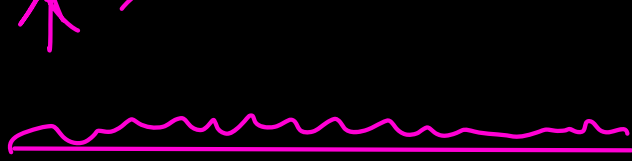
(Darcy's Equation)

$$\Delta P = \frac{4 \cdot f \cdot \rho \cdot L \cdot v^2}{2D}, Pa$$

f = friction factor,
coeff. of Resis.

ρ = Density of air





L = length of airway
(m)

V = Avg. velocity
(m/s)

D = diameter of
Airway

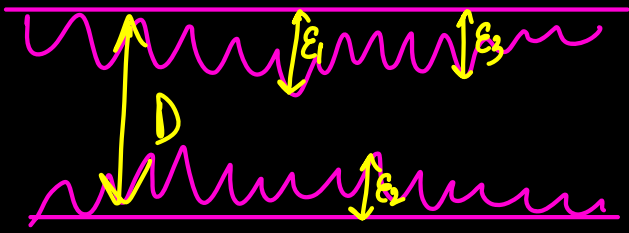
> Relationship b/w Darcy's friction factor
& Reynold's number

$$\Delta P = \frac{32 \mu L V}{D^2} = \frac{4 f 8 L V^2}{2 D}$$

$$\Rightarrow f = \frac{16 \mu}{8 V D} = \frac{16}{(8 V D / \mu)} = \frac{16}{Re}$$

$$\Rightarrow \boxed{f = 16 / Re}$$

> Relative roughness of Airway:



It is ratio of mean height of the surface irregularities to the diameter of the opening.

$$\text{Relative roughness} = \frac{\text{mean height of surface irregularities } (\varepsilon)}{\text{Diameter of the opening}}$$

> Moody's diagram : (f , RR , Re relationship)

Resistance coefficient / Friction factor (f)

In laminar flow,

Hagen-Poiseuille
Eqn

$$f = \frac{64}{Re}$$

$$\Delta P = \frac{32 \mu V L}{D^2} = \frac{64 \mu}{\sqrt{D S}} \cdot \frac{L}{D} \cdot \frac{S V^2}{2} = \frac{64}{(\sqrt{D S} \mu)} \cdot \frac{L}{D} \cdot \frac{S V^2}{2}$$

$$= \frac{64}{Re} \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

$$= \frac{f L \rho v^2}{2D}$$

For Turbulent Flow,

Blasius Eqn.

$$f = \frac{0.316}{Re^{0.25}}$$

For $Re \leq 100,000$

For $Re > 100,000$,

$$\frac{1}{\sqrt{f}} = 2.1 \log (Re \sqrt{f} - 0.8)$$

Kar

Karman - Prandtl eqn

Pressure loss in mine airways.

Total pressure loss = frictional loss + shock loss.

$$\Delta P = \underset{\downarrow}{\Delta P_f} + \underset{\downarrow}{\Delta P_s}$$

$$\Delta P_s = \Delta P - \Delta P_f$$

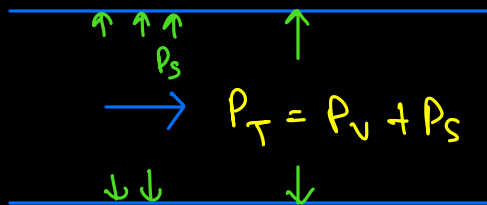
$$= \frac{K \cdot S \cdot Q^2}{A^3} + \frac{0.6x}{A^2} Q^2$$

$\Delta P_s \propto P_v$
velocity pressure.

$$= R_f Q^2 + R_s Q^2$$

$$= (R_f + R_s) Q^2$$

$$\Delta P = (R_f + R_s) Q^2 = R Q^2$$

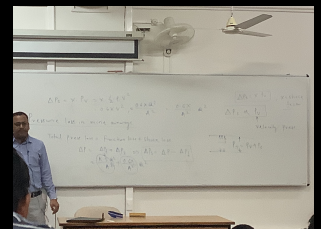


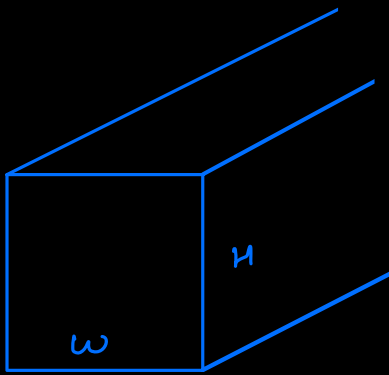
$$\therefore R = R_f + R_s$$

$$P_v = \frac{1}{2} \rho \cdot V^2$$

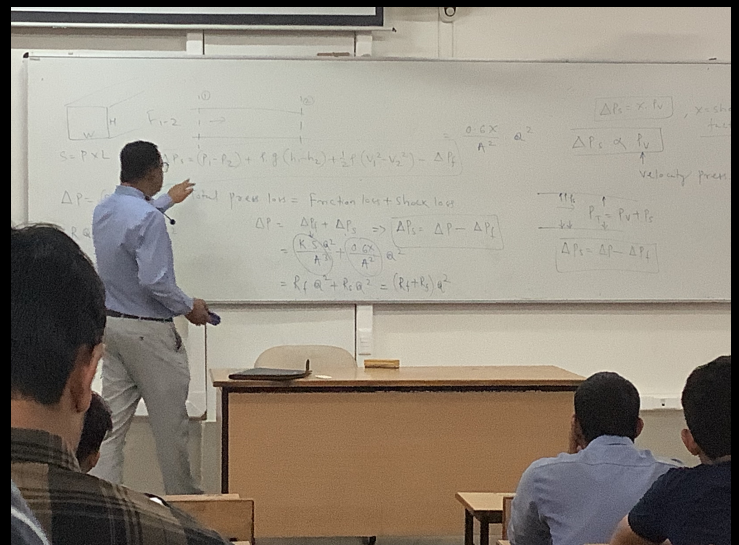
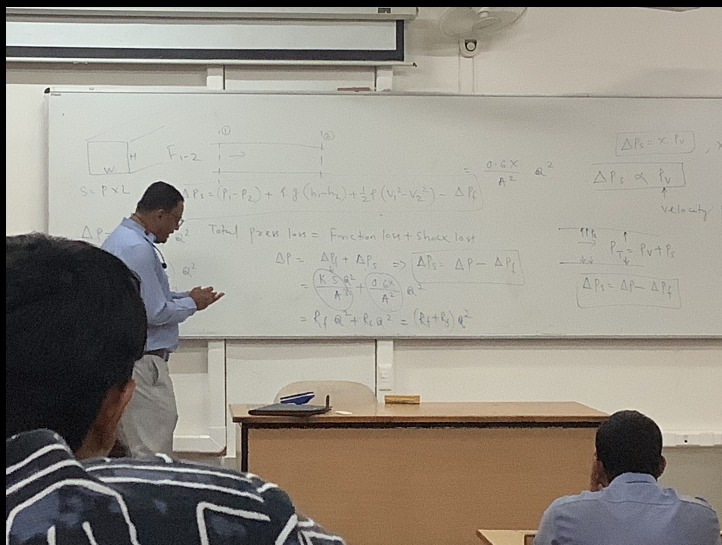
shock factor — does not have a unit.
 $\Delta P_s = x \cdot P_v$

$$\Delta P_s = x \cdot P_v = x \cdot \frac{1}{2} \cdot \rho \cdot V^2 = 0.6x V^2$$





$$S = P \times L$$

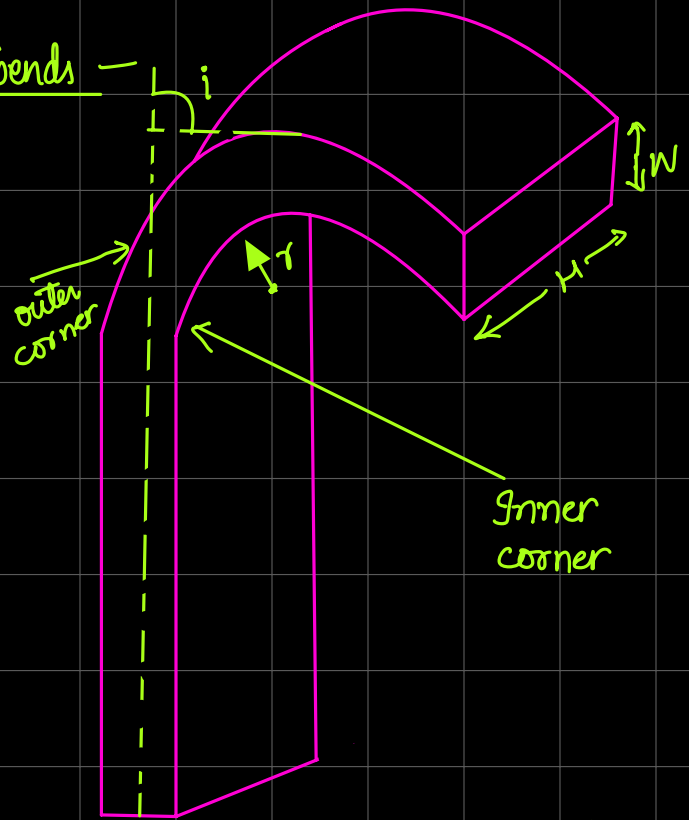


19 March 2024

★ Calculation of shock factor for bends —

Types of bends —

- ① Normal bend
- ② Square bend



characteristics of a bend

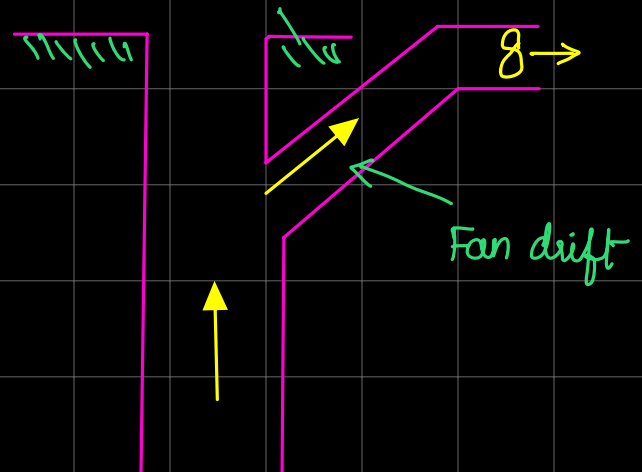
$$\Delta P = \Delta P_f + \Delta P_s$$

$$\Delta P = RQ^2$$

$$R = \frac{\Delta P}{Q^2}, \text{Ns}^2/\text{m}^8$$

$$\Delta P = \frac{K_S}{A^3} + \frac{0.6 \cdot x}{A^2} \cdot Q^2$$

$$R = R_f + R_s$$



★ Shock factor for normal bend

$$X = \frac{0.25}{R^2 \sqrt{a}} \left(\frac{2i}{\pi} \right)^2$$

For right-angled bend ; $i = \pi/2$ radians

$$X = \frac{0.25}{R^2 \sqrt{a}}$$

In case of a right-angled, square or circular opening, $X = \frac{0.25}{R^2}$

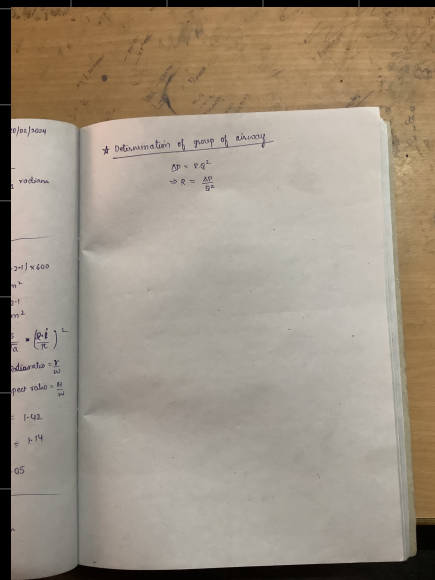
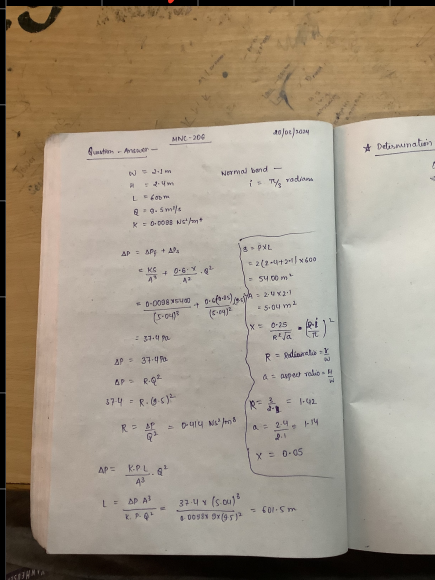
$$\text{Aspect ratio, } a = \frac{H}{W}$$

Shock factor for a square bend with rectangular opening

$$X = \frac{0.6}{R\sqrt{a}} \cdot \left(2i/\pi\right)^2$$

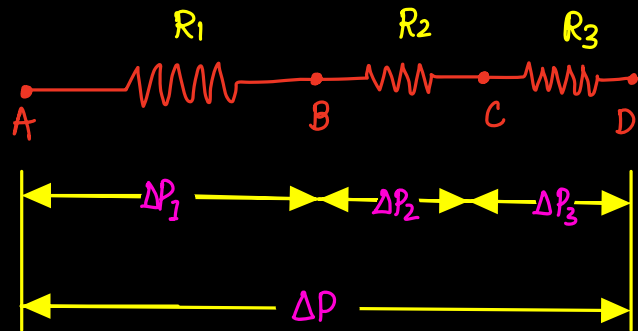
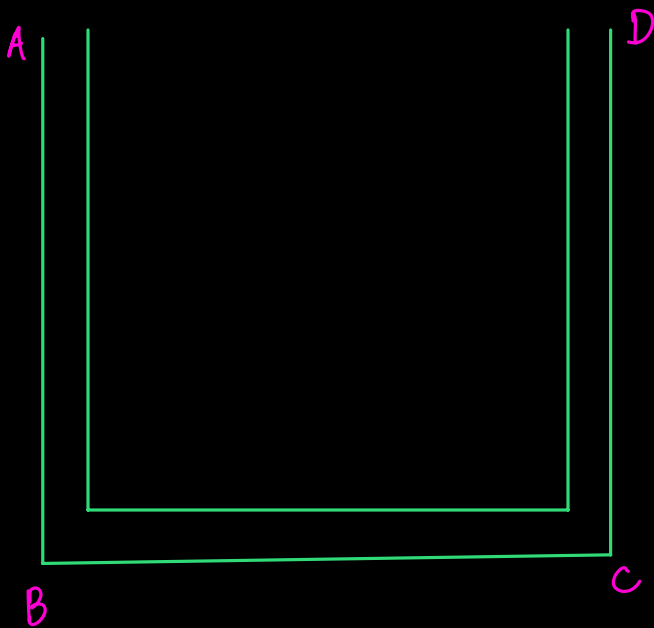
In mine airway of 2.1 m wide, 2.4 m high and 600 m long. contains a normal bend with a deflection of $\pi/3$ radian.

Radius of curvature, $r = 3$ m. The airway passes a quantity of $9.5 \text{ m}^3/\text{s}$ of air. Calculate the pressure loss in the airway as well as resistance of the airway. Also calculate the equivalent length of a straight airway of similar size & surface which will cause the same resistance. Assume the value of $k = 0.0098 \text{ N s}^2/\text{m}^4$.



★ Resistance of a group of airways ———

1. Series combination of airways



$$RQ^2 = R_1Q^2 + R_2Q^2 + R_3Q^2$$

$$\Delta P = \Delta P_1 + \Delta P_2 + \Delta P_3$$

* If there are 'n' no. of airways ———

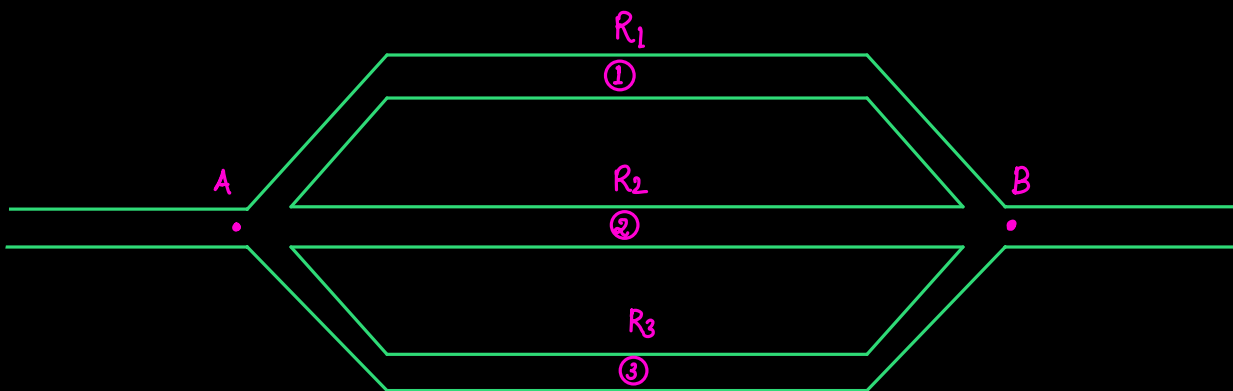
$$R = \sum_{i=1}^n R_i$$

∴ R = equivalent resistance of airways

$$RQ^2 = (R_1 + R_2 + R_3)Q^2$$

$$\Rightarrow R = (R_1 + R_2 + R_3)$$

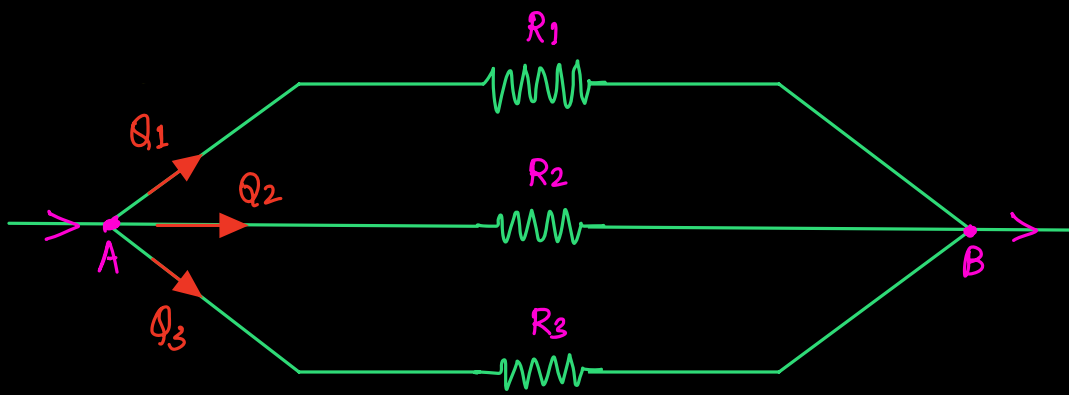
② Parallel combination of airways : —



$$Q = Q_1 + Q_2 + Q_3 \quad \text{————— ①}$$

$$\Delta P = \Delta P_1 = \Delta P_2 = \Delta P_3 \quad \text{————— ②}$$

if R = equivalent resistance



$$RQ^2 = R_1Q_1^2 = R_2Q_2^2 = R_3Q_3^2$$

$$Q_1 = Q \sqrt{\frac{R}{R_1}} = \sqrt{\frac{\Delta P}{R_1}} \quad \text{--- (3)}$$

$$Q_2 = Q \sqrt{\frac{R}{R_2}} = \sqrt{\frac{\Delta P}{R_2}} \quad \text{--- (4)}$$

$$Q_3 = Q \sqrt{\frac{R}{R_3}} = \sqrt{\frac{\Delta P}{R_3}} \quad \text{--- (5)}$$

$$\Delta P = RQ^2$$

$$\begin{aligned} Q &= \sqrt{\frac{\Delta P}{R}} = Q_1 + Q_2 + Q_3 \\ &= \sqrt{\frac{\Delta P}{R_1}} + \sqrt{\frac{\Delta P}{R_2}} + \sqrt{\frac{\Delta P}{R_3}} \end{aligned}$$

$$\boxed{\frac{1}{\sqrt{R}} = \frac{1}{\sqrt{R_1}} + \frac{1}{\sqrt{R_2}} + \frac{1}{\sqrt{R_3}}}$$

$$R = \left(\frac{1}{\frac{1}{\sqrt{R_1}} + \frac{1}{\sqrt{R_2}} + \frac{1}{\sqrt{R_3}}} \right)^2 = \left(\frac{1}{\sum_{i=1}^n \frac{1}{\sqrt{R_i}}} \right)^2$$

★ Advantages of splits in airways : —

- (i) Splits reduce the overall resistance in the mine and increase fan quantity.
- (ii) Control of quantities to different districts is done easily by controlling the resistance of the splits.
- (iii) Splitting helps in providing fresh or uncontaminated air to each district.
- (iv) Explosion of fire in districts can be easily confined.
- (v) Splitting helps in reducing the air velocities by distributing the flow.

Q.} Given five airways in series with following resistances Find the equivalent resistance and total ^{head} pressure loss with $Q = 4.719 \text{ m}^3/\text{s}$.

$$R_1 = 22.36 \text{ N s}^2/\text{m}^8$$

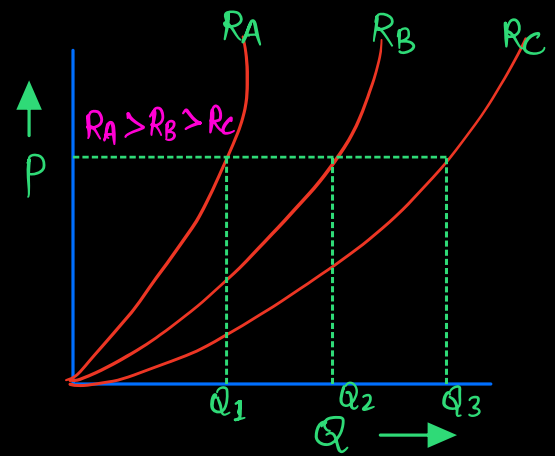
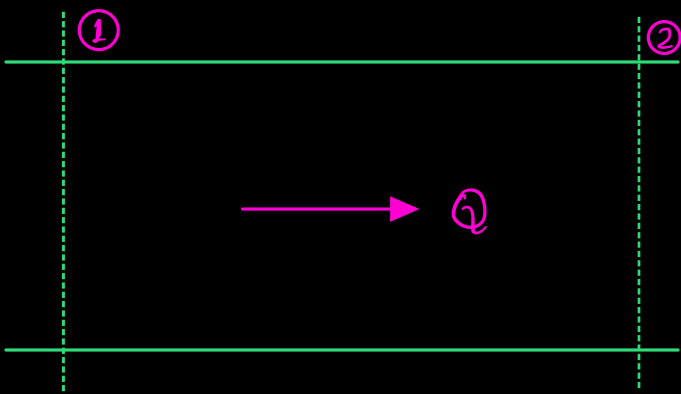
$$R_2 = 33.54 \text{ N s}^2/\text{m}^8$$

$$R_3 = 11.18 \text{ N s}^2/\text{m}^8$$

$$R_4 = 22.36 \text{ N s}^2/\text{m}^8$$

$$R_5 = 11.18 \text{ N s}^2/\text{m}^8$$

★ Air power or power of ventilation : —



$$\text{Air power} = -M \int_1^2 v \cdot dp$$

M = mass flow rate of air, kg/s

v = specific volume of air, m^3/kg

Q = Airflow rate, m^3/s

Assuming the flow to be incompressible —

$$\text{Air power} = -M \cdot v \cdot \int_1^2 dp$$

$$= -Q (P_2 - P_1)$$

$$= Q (P_1 - P_2)$$

$$= \Delta P \cdot Q \quad \text{or} \quad \Delta P \cdot Q \times 10^{-3} \text{ kW}$$

$$\Delta P = R \cdot Q^2$$

$$\text{Air power} = RQ^2 \cdot Q \times 10^{-3} \text{ kW}$$

$$= RQ^3 \times 10^{-3} \text{ kW}$$

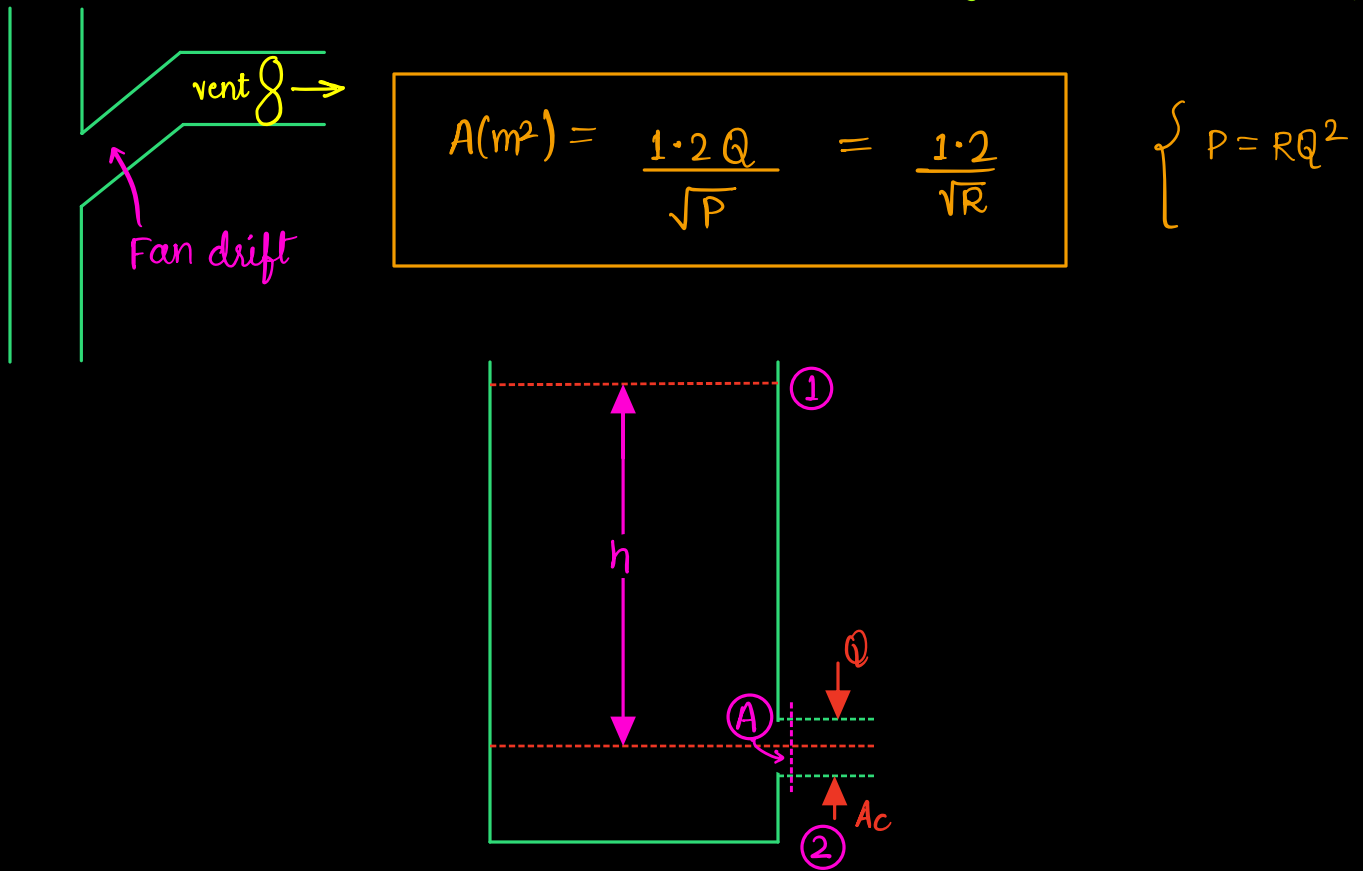
If η = efficiency of the motor & gear system of fan

$$\eta = \frac{\text{Air power}}{\text{Input power to the fan}}$$

$$\text{Input power to fan} = \frac{\text{Air power}}{\eta}$$

★ Equivalent orifice of a mine —

* Equivalent orifice of a mine is defined as the area of an imaginary opening in a thick plate which offers the same resistance to the air flow as offered by the mine itself.



A = Area of orifice

A_c = Area of vena contracta

v = velocity of flow at vena contracta, m/s = $\frac{Q}{A_c}$

$\frac{A_c}{A} = C$ = Coefficient of contraction = 0.64

$$C = \frac{A_c}{A} = 0.64 \Rightarrow \boxed{A_c = 0.64A}$$

Potential energy = velocity energy

$$\rho g h = \frac{1}{2} \rho v^2$$

$$\Rightarrow \boxed{v = \sqrt{2gh}}$$

$$\Rightarrow \frac{Q}{A_c} = \sqrt{2gh}$$

$$\Rightarrow A_c = \frac{Q}{\sqrt{2gh}} \Rightarrow 0.64 A = \frac{Q}{\sqrt{2gh}}$$

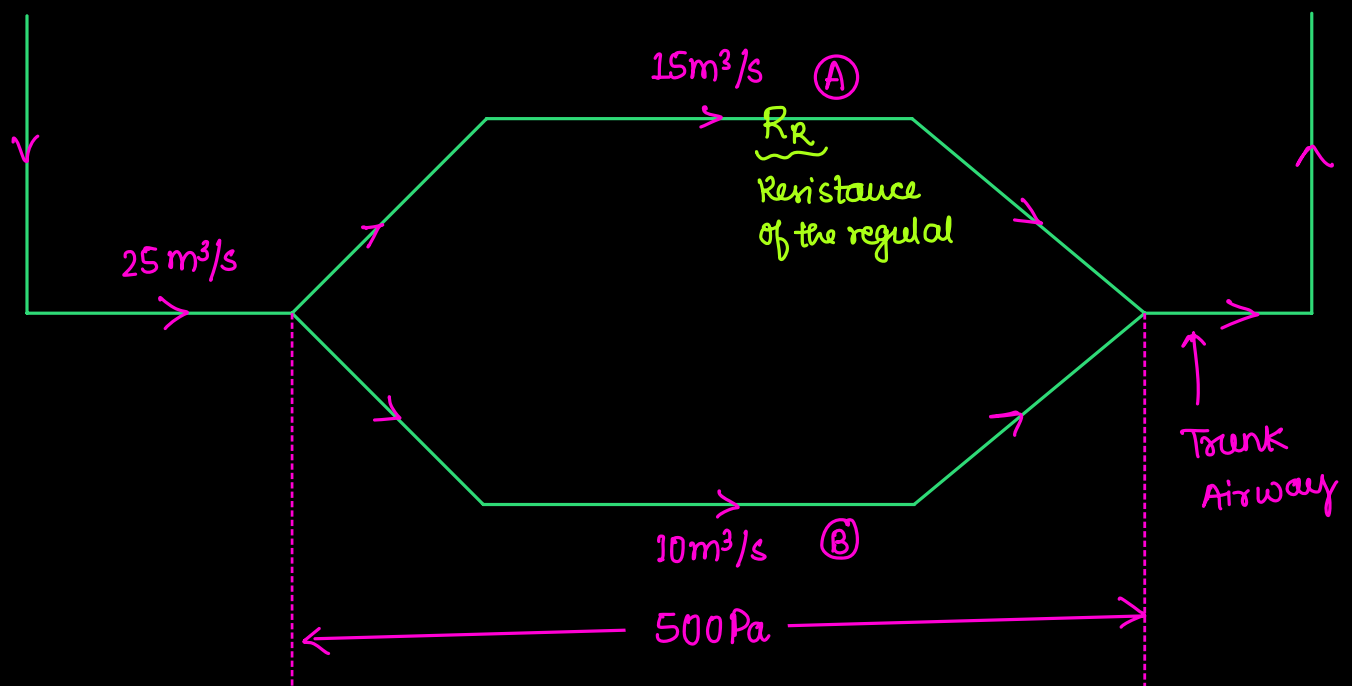
$$P = \rho g h \Rightarrow h = \frac{P}{\rho \cdot g}$$

$$\Rightarrow 0.64 A = \frac{Q}{\sqrt{2g \cdot \frac{P}{\rho \cdot g}}} = \frac{Q \cdot \sqrt{\rho}}{\sqrt{2P}}$$

* Standard density of air = 1.2 kg/m^3

$$A = \frac{Q \cdot \sqrt{\rho}}{0.64 \sqrt{2P}} = \frac{1.2 Q}{\sqrt{P}} = \frac{1.2}{\sqrt{R}}$$

Q. > Two parallel splits A & B have a pressure of 500 Pa acting across them causing flow of $15 \text{ m}^3/\text{s}$ in split A & $10 \text{ m}^3/\text{s}$ in split B. It is designed to reduce the quantity in split A to $10 \text{ m}^3/\text{sec}$. by placing a regulation in it. Calculate the quantity in split B. After installation of the regulator as well as the size of the regulator if surface total pressure is 1 kPa.



Pressure acting in Trunk airway = $1000 - 500 = 500 \text{ Pa}$

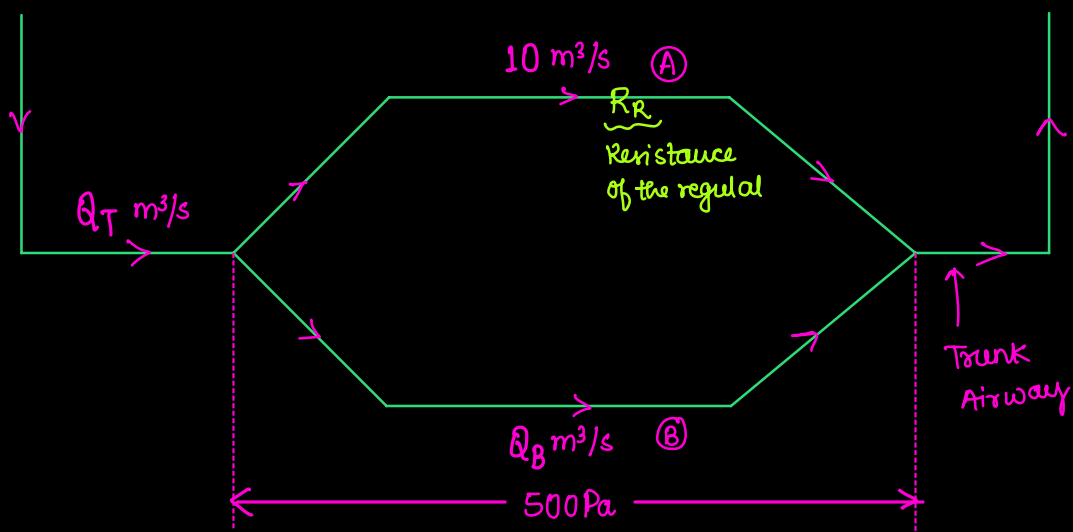
In split A : —

$$500 = R_A \cdot Q_A^2$$

$$\Rightarrow R_A = \frac{500}{15^2} = 2.2 \text{ N s}^2/\text{m}^8$$

$$R_B = \frac{500}{10^2} = 5 \text{ N s}^2/\text{m}^8$$

$$R_T = \frac{500}{25^2} = 0.8 \text{ N s}^2/\text{m}^8$$



$$Q_T = Q_A + Q_B$$

$$Q_T = 10 + Q_B$$

$$Q_B = Q_T - 10$$

After installation of regulator ———

Considering split B :

$$1000 = R_T \cdot Q_T^2 + R_B \times Q_B^2$$

$$1000 = 0.8 \times Q_T^2 + R_B (Q_T - 10)^2$$

$$1000 = 0.8 \times Q_T^2 + 5 (Q_T^2 + 100 - 20Q_T)$$

$$1000 = 0.8 Q_T^2 + 5 Q_T^2 + 500 - 100 Q_T$$

$$1000 = 5.8 Q_T^2 - 100 Q_T + 500$$

$$5.8 Q_T^2 - 100 Q_T - 500 = 0$$

$$Q_T = 21.3 \text{ m}^3/\text{s}$$

$$Q_B = Q_T - Q_A$$

$$= 21.3 - 10$$

$$= 11.3 \text{ m}^3/\text{s}$$

$$1000 = R_T \cdot Q_T^2 + (R_A + R_R) \cdot Q_A^2$$

$$1000 = 0.8 \times (21.3)^2 + (2.2 + R_R) \cdot (10)^2$$

$$R_R = 4.15 \text{ N s}^2/\text{m}^8$$

$$\text{Size of the regulator, } A = \frac{1.2}{\sqrt{R_R}} = \frac{1.2}{\sqrt{4.15}} = 0.59$$